Innovative Product Development

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Title: Learnova — AI-powered redesign of outdated lecture slides into interactive, illustrated teaching decks

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IDEA/APPROACH DETAILS

❖ Problem Statement

Lecture slides used in colleges are mostly old PowerPoint files or PDFs: dense bullet text, equations and diagrams pasted as pictures, missing titles and no interaction. Faculty do not have the time to redesign every deck, and AI deck generators create new slides from a prompt but rewrite or drop the teacher's content and cannot redesign an existing deck faithfully.

Learnova redesigns an existing lecture deck into a modern, illustrated, interactive deck while keeping every point the teacher wrote.

❖ Objectives

1. Read real lecture material (PPTX, slide PDFs, Word notes, typed topics) and recover titles, bullet levels, tables, figures and picture-only slides.
2. Choose a suitable visual for each slide (process, timeline, comparison, definition, chart …) and draw an illustration on every slide.
3. Keep 100% of the source content: no item dropped, no fact invented, no word broken across lines.
4. Make slides interactive: checkpoint quizzes, step-by-step solved problems and hands-on labs for hard topics (NLP, ML, IPCV).
5. Work with AI keys for better titles, explanations and diagrams, and fully offline when keys are unavailable.

Proposed Solution — Learnova

One upload in, a complete redesigned teaching kit out — faithful to the teacher's content

1 Read

- PPTX, slide PDF, Word notes or a typed topic
- Titles, bullet levels, tables, figures recovered
- OCR for picture-only slides; logos, footers, page numbers removed
- Original colour palette carried into the new design

2 Redesign

- Visual chosen per slide: process, timeline, comparison, definition, chart, cards …
- Assertion titles that state the point
- Overloaded slides split, thin slides merged
- Text fitted glyph by glyph — never overflows or breaks words

3 Illustrate

- An illustration on every topic slide
- AI-written diagram of the slide's idea (Gemini, 5-key pool)
- Teacher's own figure kept beside the text
- Offline illustrator when no key — each linked as an HTML page

4 Engage

- Checkpoint quizzes from the content
- Exact step-by-step solved problems (Bayes, Viterbi, CYK …)
- Interactive labs: Region Growing, Split & Merge, NLP labs
- Flip cards, matching activity, deck tutor

5 Deliver

- Editable PowerPoint with native shapes
- Interactive web deck (Reveal.js)
- Study pack: flashcards, Anki, quiz bank, cheat sheet
- Share link, presenter console, mobile app (PWA)

Innovation & uniqueness: redesigns the deck a teacher already has instead of generating a new one; every structural decision is an explainable rule; AI is used only where its output can be checked (titles, explanations, diagrams, labs) and the system keeps working fully offline.

Literature Review/Existing Systems

Sr. No.	Reference Paper / Existing Application	Key Findings	Gap / Scope for Improvement
1	Hu & Wan, “PPSGen: Learning-based presentation slides generation for academic papers”, IEEE TKDE, 2015	Regression model scores sentence importance; ILP selects sentences into slides.	Works on papers only; summarises (loses content); no visuals, no redesign of existing slides.
2	Fu et al., “DOC2PPT: Automatic presentation slides generation from scientific documents”, AAAI, 2022	Neural summariser + figure selection + layout prediction, trained on paper–slide pairs.	Needs training data; abstractive text can drop or alter facts; input must be a paper.
3	Sun et al., “D2S: Document-to-slide generation via query-based text summarization”, NAACL, 2021	Slide titles used as queries; QA-style retrieval + summarisation per slide.	Text only; no figures or interactivity; cannot rebuild an existing deck.
4	Garner & Alley, “How the design of presentation slides affects audience comprehension: assertion–evidence approach”, IJEE, 2013	Sentence headlines + visual evidence improve comprehension over bullet lists.	Design guidance only — no tool applies it automatically to old decks.
5	Existing tools: Gamma, Microsoft Copilot in PowerPoint, Gemini in Slides, Canva Magic Design	LLM outline → template fill; good-looking new decks from a prompt.	Rewrite/merge the teacher's content, lose original figures, no quizzes/labs, need internet and paid AI.

Common limitations: existing systems generate new decks instead of redesigning old ones, are not faithful to the source, ignore picture-only and equation slides, and offer no interactivity or offline mode.

Technical Approach & Methodology

Processing workflow with the algorithm used at each stage

1 Upload & convert

PPTX → PDF (LibreOffice), DOCX, typed text → Markdown IR

2 Slide reader

Line geometry; furniture = text repeated on ≥ max(3, 0.3n) pages; bullet level = nearest x-anchor

3 OCR decision

Graphic ratio $\gamma = |\text{ink} \setminus \text{text}| / |\text{ink}|$; use OCR only if conf ≥ 62, $\gamma \le 0.42$, little symbol debris

4 Planner

Split slides over the density budget; merge thin parts; agenda, glossary, quizzes

5 Decision engine

27 detectors → (family, confidence, data, reason); $d^* = \text{argmax}$ confidence after exclusion rules

6 AI assist (optional)

Assertion titles, explanations, key terms — accepted only if grounded in the slide's words

7 Illustrations

LLM writes SVG → validated (safe elements, ≥ 6 shapes, renders) → cached; else offline scene

8 Layout & build

Font s^* = largest size that fits; word scale $\sigma = \min(1, W / w(\text{longest word}))$; PPTX + web deck

9 Audit

Overflow, broken words, sparse slides; fact retention $R = |F_{src} \cap F_{deck}| / |F_{src}|$

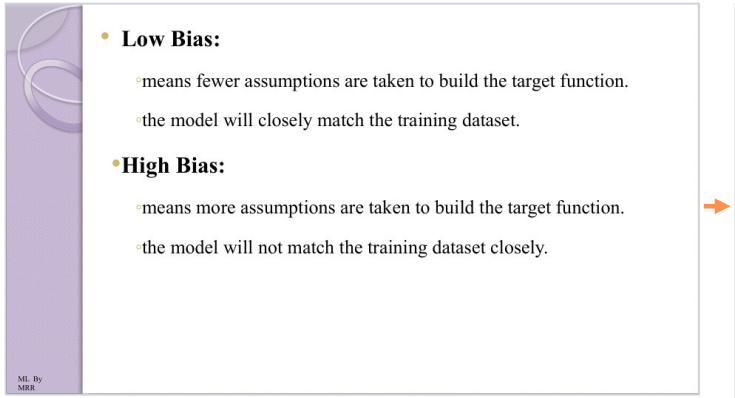
IDEA

Prototype — upload an old deck, get a redesigned interactive deck

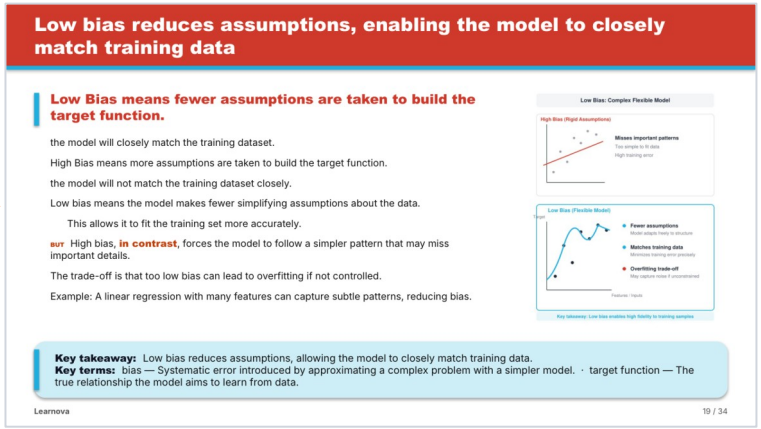
How it works

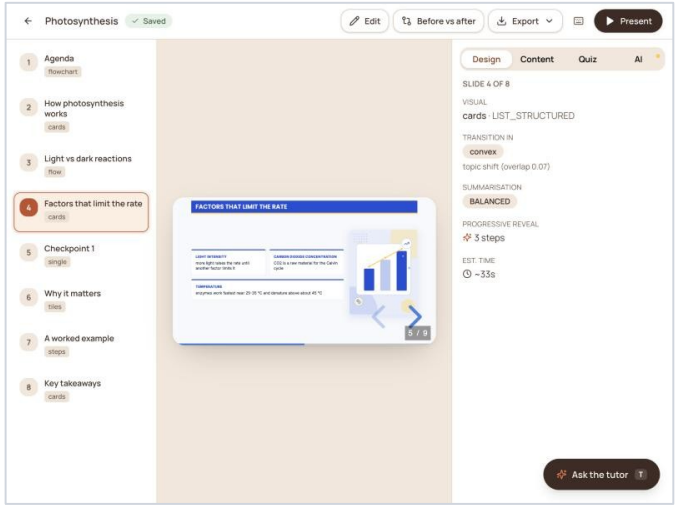
1. Upload a lecture deck — PPTX, PDF, Word notes — or type a topic.
2. Learnova reads every slide: titles, bullet levels, tables, figures; OCR for picture slides.
3. Each slide gets the right visual, an assertion title, and an illustration (AI-drawn diagram, the teacher's figure, or an offline illustration).
4. Quizzes, step-by-step solved problems and labs are added.
5. Download an editable PowerPoint, an interactive web deck and a study pack; share a link.

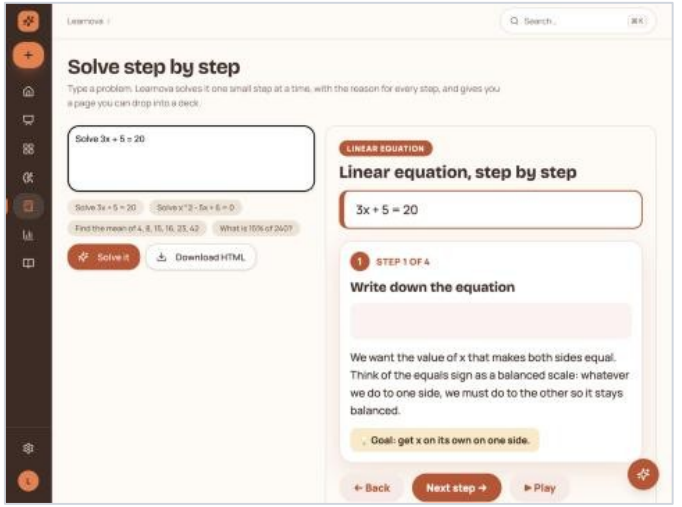
Before — original slide



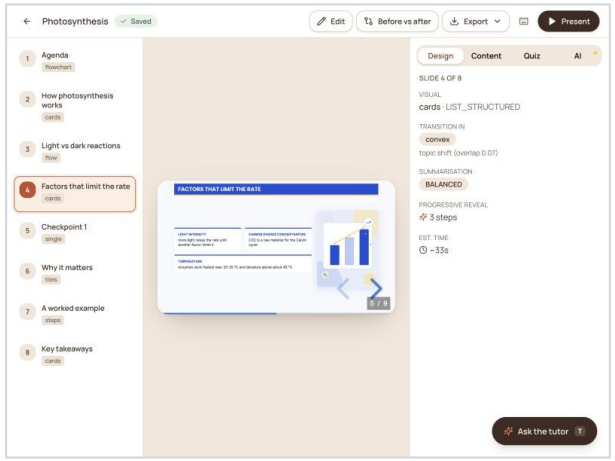
After — Learnova slide



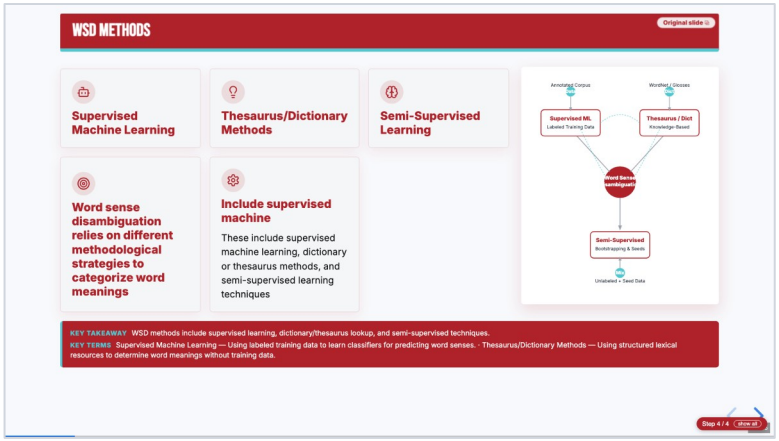




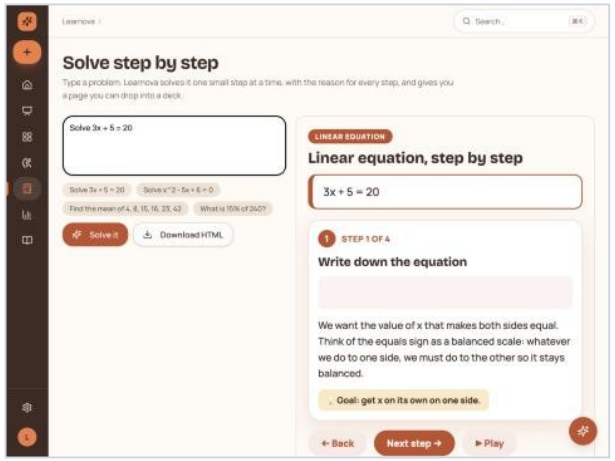
Prototype — working application screens



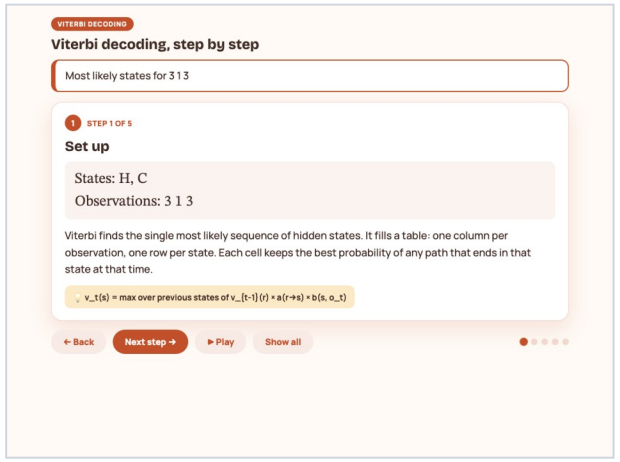
Deck preview: every slide with the visual chosen and why



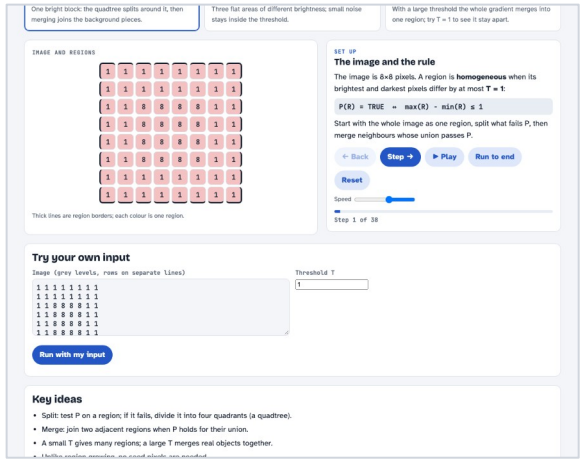
Interactive web deck with AI diagram



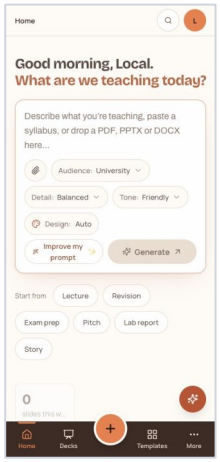
Solve step by step: beginner-level walkthrough



Viterbi explainer, computed exactly



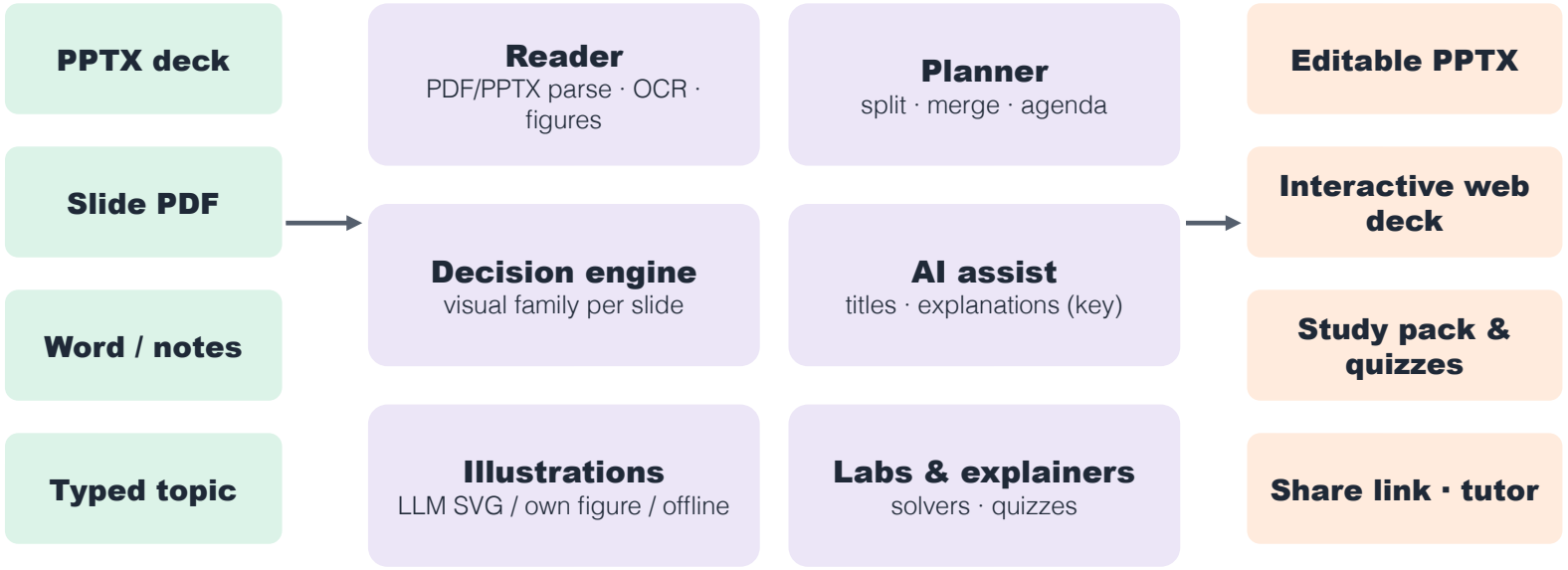
Region Splitting & Merging lab (IPCV)



Phone layout / installable app

ARCHITECTURE

Input → Processing (workflow) → **Output**



Layout engine + audit: text measured glyph by glyph — no overflow, no broken words

FastAPI service · job queue · deck library · OTP sign-in · AI router (Gemini ×5 → Groq → NVIDIA → offline rules)
React front-end: Create · Generate (live progress) · Preview · Present · Solve · Export · Settings

TECHNOLOGY STACK

- **Front-end:** React 18, Vite, Tailwind CSS, Reveal.js (web deck), installable PWA
- **Back-end:** Python 3.12, FastAPI, background job queue, SQLite
- **Document processing:** python-pptx, PyMuPDF, LibreOffice (PPTX→PDF), Tesseract OCR, python-docx
- **AI / NLP:** Google Gemini (5-key pool), Groq, NVIDIA NIM — router with failover; rule-based offline engine
- **Maths & labs:** SymPy (exact solvers), vanilla JS lab engine, SVG illustrations
- **Quality:** pytest (750+ tests), layout audit, gold-set evaluation

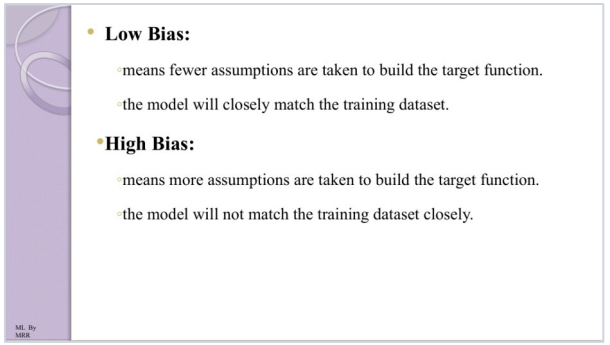
Before vs After — problems faced → what is achieved

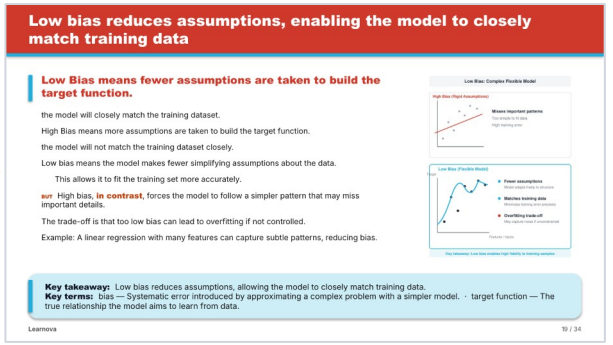
Area	BEFORE — problem faced	AFTER — with Learnova
Text-heavy slides	✘ Walls of bullets, 100+ words on one slide; students skim and miss points	✔ One idea per slide drawn as a visual (process, timeline, table, definition, cards); split at a readable density
Pictures of equations & diagrams	✘ Content locked in images; nothing searchable; tools drop these slides	✔ Every picture slide kept as a figure; OCR only when the text is reliable; derivations shown full size
Untitled / generic titles	✘ “Slide 9”, “Hypothesis (2/5)” — the title says nothing	✔ Assertion titles that state the point; split parts named after what they cover
No visuals for many topics	✘ Plain text with no diagram to explain the idea	✔ An illustration on every topic slide: AI-drawn diagram, teacher's figure or offline scene — with an HTML link
No interaction	✘ Students only read; solved problems shown as finished tables	✔ Checkpoint quizzes, exact step-by-step explainers, labs (Region Growing, Split & Merge, CYK, Viterbi, NB-WSD)
AI tools change the content	✘ Generators paraphrase, merge items, invent facts; nothing can be checked	✔ Content kept verbatim and checked: 98–100% of facts retained, explanation for every design choice
Redesign takes hours	✘ Faculty spend an evening per lecture; needs design skill	✔ Upload → finished PPTX + web deck in ~10–60 s; fully editable
Needs internet / paid AI	✘ Cloud-only tools, subscriptions, quotas	✔ Works offline; free API keys used only to improve; automatic fallback when a key runs out

Before vs After — the same slides, redesigned

Left: original lecture slide · Right: slide generated by Learnova from it

ML · Bias and variance (PPTX)

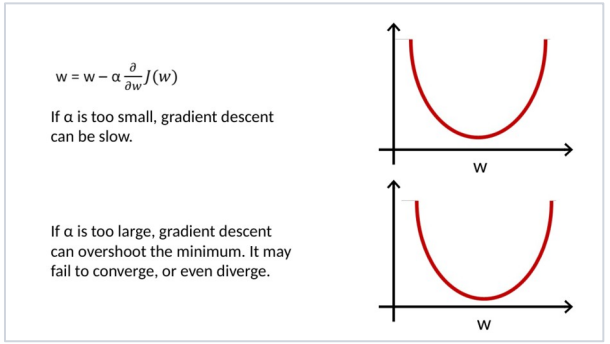


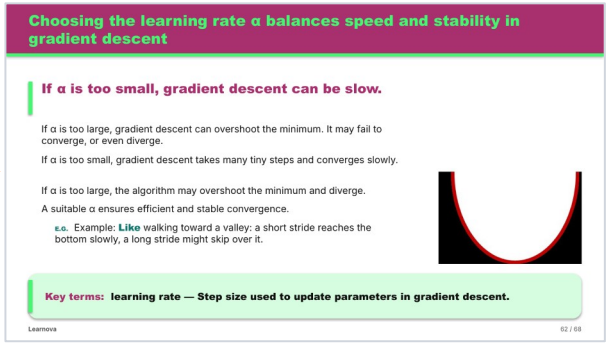


What changed

Untitled bullet slide → assertion title, full definitions, the teacher's diagram beside the text, key takeaway and key terms.

ML · Linear regression (PPTX)

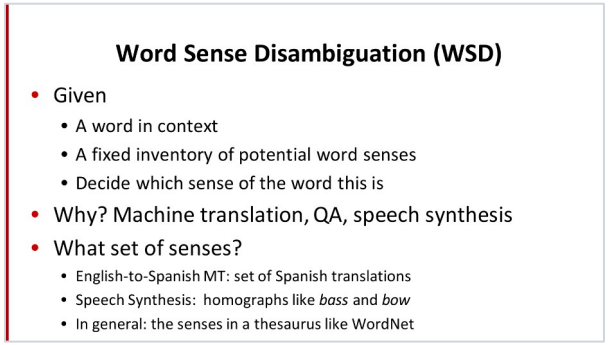


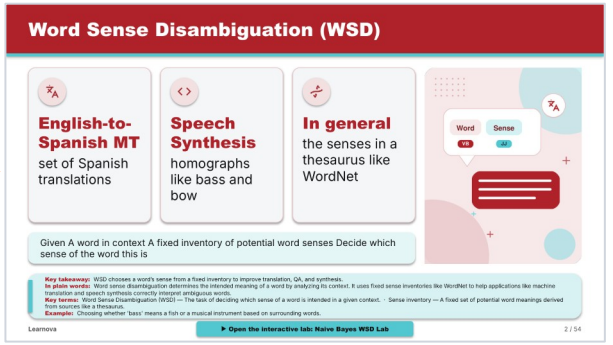


What changed

Picture-only slide of plots → kept as a figure with its explanation as text; title states the learning-rate trade-off.

NLP · Word sense disambiguation (PDF)





What changed

Dense bullet list → structured cards with an AI-drawn diagram, quiz and a Naive Bayes WSD lab link.

Before vs After — measured on our test decks

Original deck → Learnova deck (offline rules + Gemini keys)

Deck	Slides	Untitled	Slides with picture	Visual structures	Quizzes	Labs	Linked illustrations	Fact retention
ML Bias & Variance (PPTX)	28 → 34	14 → 0	22 → 29	0 → 22	0 → 5	0 → 2	0 → 12	100%
ML Linear Regression (PPTX)	63 → 68	36 → 0	49 → 64	0 → 45	0 → 4	0 → 8	0 → 23	100%
WSD lecture slides (PDF)	44 → 54	1 → 0	7 → 46	0 → 52	0 → 7	0 → 10	0 → 40	98%
NLP Module 4 (PDF)	36 → 49	0 → 0	13 → 43	0 → 43	0 → 6	0 → 3	0 → 37	100%
NLP Module 5 (PDF)	95 → 102	32 → 0	49 → 96	0 → 89	0 → 6	0 → 13	0 → 78	99%

83 → 0
untitled slides

251
slides drawn as a visual

28
checkpoint quizzes added

36
interactive lab links

190
illustrations with HTML link

98–100%
facts kept

USE CASES

Where Learnova is used

Faculty before a lecture

Uploads last year's PPT; gets a redesigned, illustrated deck with quizzes in under a minute, fully editable in PowerPoint.

Students revising for exams

Open the web deck, flip key-term cards, solve checkpoint questions and step through solved problems at their own pace.

Lab-heavy subjects

IPCV / NLP / ML topics get hands-on labs: region growing, split & merge, CYK, Viterbi, Naive Bayes WSD, HMM tagging.

Colleges without paid AI

Everything works offline with rule-based logic; AI keys only improve titles, explanations and diagrams.

Remote and flipped classes

Share a view-only link; the deck tutor answers questions from the deck's own content.

Accessibility

Large readable text, no broken words, plain-language explanations and alt text for every illustration.

Implementation

Modules developed and integrated

Module	What is implemented	Status
Input readers	PPTX via LibreOffice → slide reader; PDF page reader with OCR; DOCX; typed topics	Done
Content Decision Engine	27 detectors, 20+ visual families, reason for every choice	Done
PowerPoint + web builders	Native editable shapes; Reveal.js deck with click-by-click reveals	Done
Illustrations	AI SVG diagram per slide (5 Gemini keys) · source figure · offline scenes; HTML link	Done
Quizzes & explainers	Checkpoint MCQs; exact solvers incl. Bayes, bigram, Naive Bayes, Viterbi, CYK	Done
Interactive labs	7 built-in (incl. Region Growing, Split & Merge) + generated labs per topic	Done
Platform	FastAPI jobs, OTP login, library, share links, study pack, PWA front-end	Done
Evaluation & papers	Gold-set accuracy, layout audit, fact retention; IEEE + Springer papers	Done
Classroom study	Controlled study with students on learning outcomes	Planned

Feasibility & Viability

Challenges we met, how they are handled, and what it costs to run

Challenge	Why it is hard	How Learnova handles it
Real lecture PDFs are messy	Symbol-font bullets, glued titles, picture-only slides, repeated logos	Page reader with OCR decision, title segmentation by dynamic programming, furniture removal
AI errors / hallucination	LLMs paraphrase, drop items, miscalculate	AI output accepted only if grounded in the slide; numbers computed by SymPy; source text kept verbatim
API quota and cost	Free tiers limit requests; keys get exhausted	5-key Gemini pool with per-key pacing, Groq/NVIDIA failover, full offline mode, response cache
Layout breaking in PowerPoint	Overflow, words split across lines, tiny fonts	Glyph-measured fitting, minimum 11 pt then continuation slide, automatic layout audit
Speed for large decks	216-page decks, many diagrams and labs	Parallel generation (labs, diagrams), caching by content hash: 216 pp in ~35 s offline
Unsafe generated code	AI-written labs and SVG could contain scripts	Whitelisted SVG elements; lab JS checked with node --check, no network or storage APIs

Viability: runs on a college laptop or one small server (Python + LibreOffice + Tesseract); free-tier AI keys are optional; open formats (PPTX, HTML) need no special viewer; freemium model possible — free offline tier, paid AI diagrams and LMS integration.

Impact & Benefits

Students

- ✔ Clear slides: one idea per slide with a picture
- ✔ Solve problems step by step with reasons
- ✔ Practice with quizzes, labs and flashcards
- ✔ Revise on phone, offline

Faculty

- ✔ Old decks modernised in about a minute
- ✔ Content kept exactly — nothing to re-check
- ✔ Editable PowerPoint, ready to present
- ✔ Quizzes and labs without extra preparation

Institution

- ✔ Consistent, accessible course material
- ✔ No paid AI licence required
- ✔ Supports flipped and remote classes
- ✔ Reusable labs library across subjects

Society (SDG 4)

- ✔ Quality education with low-cost tools
- ✔ Works where internet is unreliable
- ✔ Plain-language explanations help first-generation learners
- ✔ Open web formats, easy to share

Measured so far: on seven real course decks every slide was kept, untitled slides went to zero, every topic slide received an illustration, and quizzes and labs were added automatically. A classroom study on learning outcomes is planned for the next semester.

Novelty — what Learnova does that others do not

Capability	Learnova	Gamma	Copilot PPT	Gemini Slides	Canva
Redesigns an existing deck (not only new ones)	✔	✘	~	~	~
Keeps every source point (fact check)	✔	✘	✘	✘	✘
Keeps teacher's figures, OCR of picture slides	✔	✘	~	✘	✘
Illustration on every slide + HTML link	✔	~	~	~	~
Exact step-by-step solved problems	✔	✘	✘	✘	✘
Interactive labs for hard topics	✔	✘	✘	✘	✘
Works offline without AI keys	✔	✘	✘	✘	✘
Explains every design choice	✔	✘	✘	✘	✘

✔ supported ~ partly (e.g. generic stock images, or only with a paid plan / internet) ✘ not supported — based on the team's comparison of each tool's documented features (Oct 2026).

Results on real course material

97%
visual chosen correctly
101 gold slides, 3 held-out sets

98–100%
facts kept
6 real decks, 36–216 pages

0
broken words / overflow
layout audit, all decks

26 / 26
problems solved correctly
held-out explainer benchmark

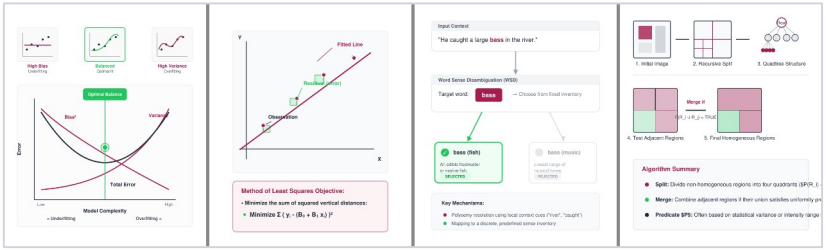
< 1 ms
decision per slide
rule-based engine

~35 s
216-page PDF → full deck
without any API key

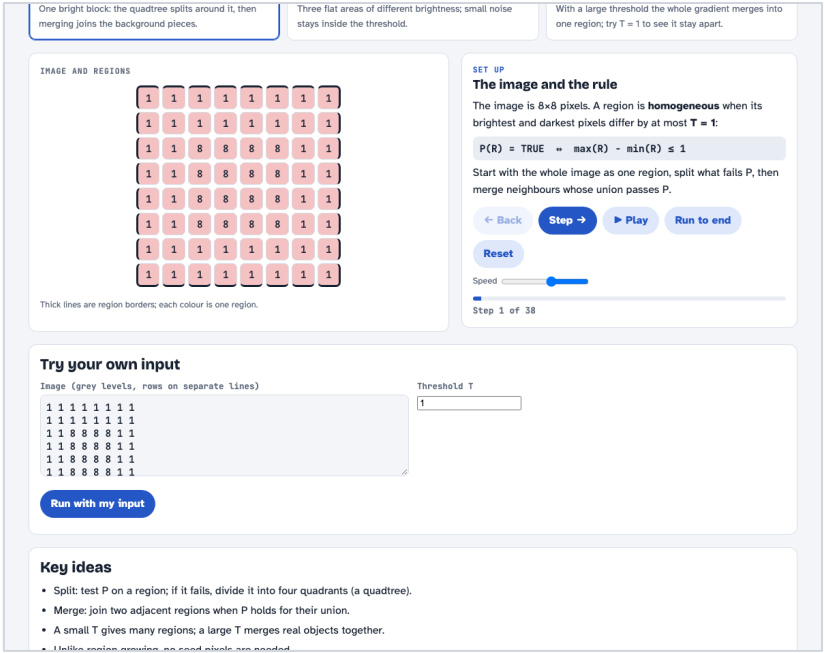
Tested on: NLP Module 4 (36 pp), Module 5 (95 pp), Syntax analysis (216 pp), WSD slides and notes, Lesk algorithm notes, ML Bias & Variance (PPTX, 28 slides) and Linear Regression (PPTX, 63 slides).

Linear Regression PPTX: earlier parser kept 23 of 63 slides; the new reader keeps all of them with the original derivations as figures.

AI-drawn diagrams (one per slide)



Region Splitting & Merging lab (IPCV)



Future scope and conclusion

Future scope

- Classroom study with students to measure learning gains.
- Narrated video export of each deck (text-to-speech).
- More labs: CNN filters, decision trees, gradient descent.
- Teacher review mode: approve or edit AI suggestions slide by slide.
- Hindi and regional-language decks.
- College LMS integration (Moodle / Google Classroom).

Conclusion

Learnova turns the decks teachers already have into clear, illustrated and interactive lessons without losing their content.

It is faithful by design (every point kept and checked), visual on every slide, interactive through quizzes, solved problems and labs, and dependable — it works with AI keys and without them.

Two research papers (IEEE conference format and Springer LNCS) describe the methods and the evaluation.

References:

Ref. Papers:

[1] Y. Hu and X. Wan, “PPSGen: Learning-based presentation slides generation for academic papers,” IEEE Trans. Knowl. Data Eng., vol. 27, no. 4, pp. 1085–1097, 2015.

[2] T.-J. Fu, W. Y. Wang, D. McDuff and Y. Song, “DOC2PPT: Automatic presentation slides generation from scientific documents,” in Proc. AAAI, vol. 36, pp. 634–642, 2022.

[3] E. Sun, Y. Hou, D. Wang, Y. Zhang and N. X. R. Wang, “D2S: Document-to-slide generation via query-based text summarization,” in Proc. NAACL, pp. 1405–1418, 2021.

[4] J. K. Garner and M. P. Alley, “How the design of presentation slides affects audience comprehension,” Int. J. Eng. Educ., vol. 29, no. 6, pp. 1564–1579, 2013.

[5] R. E. Mayer, Multimedia Learning, 2nd ed. Cambridge University Press, 2009.

[6] J. Sweller, “Cognitive load during problem solving: Effects on learning,” Cognitive Science, vol. 12, no. 2, pp. 257–285, 1988.

[7] J. Maynez, S. Narayan, B. Bohnet and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in Proc. ACL, pp. 1906–1919, 2020.

[8] R. Smith, “An overview of the Tesseract OCR engine,” in Proc. ICDAR, pp. 629–633, 2007.

[9] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed. draft, 2025.

References Web resources:

Gamma — https://gamma.app · Microsoft Copilot in PowerPoint · Google Gemini in Slides · Canva Magic Design — accessed Oct 2026

Reveal.js — https://revealjs.com · python-pptx — https://python-pptx.readthedocs.io · PyMuPDF — https://pymupdf.readthedocs.io

Project repository — https://github.com/SDP42/LearnovaV2